

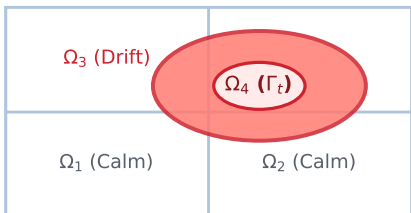
DC-JSR: Defect-Constrained Stateful Schwarz Preconditioner Maintenance

Autonomous subset selection via cumulative algebraic variation: $S_t^* = \{i: V_i^-(t) > \varepsilon\}$ with coupled Galerkin coarse synchronization

Stage 1: Evolving Operator

Sparse SPD System Sequence

$$A_t x_t = b_t, \quad t = 1, \dots, T$$



- Local operator: $A_i(t) = R_i A_t R_i^T$
- Operator drift: $\Delta A_t = A_t - A_{t-1}$
- Pronounced spatial locality:
 - Minor subset undergoes intense drift
 - Broad background is quasi-static
- Full Rebuild: 25%–80% waste
- Static Reuse: iteration explosion

Stage 2: Persistent State

$\mathcal{O}(n_i)$ Diagonal Sensing

$$\mathbf{d}_i(t) = \text{diag}(A_i(t))$$

$$\mathbf{v}_i(t) = \frac{|\mathbf{d}_i(t) - \mathbf{d}_i(t-1)|_2}{|\mathbf{d}_i(t)|_2}$$

Extract
Diag
→

Accumulated State-State Proxy:

$$V_i^-(t) = V_i(t-1) + \mathbf{v}_i(t)$$

- Provenance interval: $[\tau_i, t]$
- $$V_i^-(t) = \sum_{s=\tau_i+1}^t \mathbf{v}_i(s)$$
- Normalized diagonal endpoint bound:

$$\frac{|\mathbf{d}_i(t) - \mathbf{d}_i(\tau_i)|_2}{|\mathbf{d}_i(t)|_2} \leq \gamma_i(t) V_i^-(t)$$
- Replaces heuristic age penalty
- Algebraic proxy, not spectral norm
- Persistent tuple: $\mathcal{M}_t = (\{K_i\}, \{\tau_i\}, A_0, \{V_i\})$

Evaluate
Proxy
→

Stage 3: Subset Selection

Defect-Constrained Problem

$$\begin{aligned} \min_{S \subseteq \{1, \dots, M\}} |S| \\ \text{s. t. } \max_{j \notin S} V_j^-(t) \leq \varepsilon \end{aligned}$$

Unique Closed-Form $\mathcal{O}(M)$ Solution:

$$S_t^* = \{i: V_i^-(t) > \varepsilon\}$$

Active Refactor: $i \in S_t^*$

- In-place numeric refactor: $A_i(t)$
- Symbolic METIS structure reused
- Reset proxy: $V_i(t) \leftarrow 0$
- Advance provenance: $\tau_i \leftarrow t$

Factor Reuse: $j \notin S_t^*$

- Zero setup overhead
- Retain cached inverse: $A_j(\tau_j)^{-1}$
- Retain variation: $V_j(t) \leftarrow V_j^-(t)$
- Bounded drift: $V_j(t) \leq \varepsilon$

Stage 4: Preconditioned Solve

Two-Level Preconditioner Q_t

$$\begin{aligned} Q_t = & \sum_{i \in S_t^*} R_i^T D_i A_i(t)^{-1} D_i R_i \\ & + \sum_{j \notin S_t^*} R_j^T D_j A_j(\tau_j)^{-1} D_j R_j \\ & + Z A_0(t)^{-1} Z^T \end{aligned}$$

Coarse Synchronization:

$$A_0(t) = Z^T A_t Z \in \mathbb{R}^{M \times M}$$

Prevents low-frequency error drift

Certified PCG Solve

- Dual independent residual checks
- $\frac{\|b_t - A_t x_t\|_2}{\|b_t\|_2} \leq 3.28 \times 10^{-9} \ll 10^{-8}$
- Zero solver retries across 876 solves
- Up to 25.9% wall-clock net speedup
- Shock absorption at multi-front & jumps

Core Insight: Tracking path variation V_i^- along unrefreshed intervals $[\tau_i, t]$ enables autonomous transitions between zero refresh ($K=0$), partial maintenance ($K \ll M$), and emergency full rebuild ($K=M$).